

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree
in
Information Technology
(Specialization: Data Science)

Final Examination
Year 3, Semester 1 (2018)

IT3021 – Data Warehousing and Business
Intelligence

Duration: 2 Hours

Instructions to Candidates:

- ◆ This paper has 5 questions. Answer all questions.
- ◆ This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence.
- ◆ Write answers in the booklet given.
- ◆ Total marks 100.
- ◆ This paper contains 6 pages including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1**(18 Marks)**

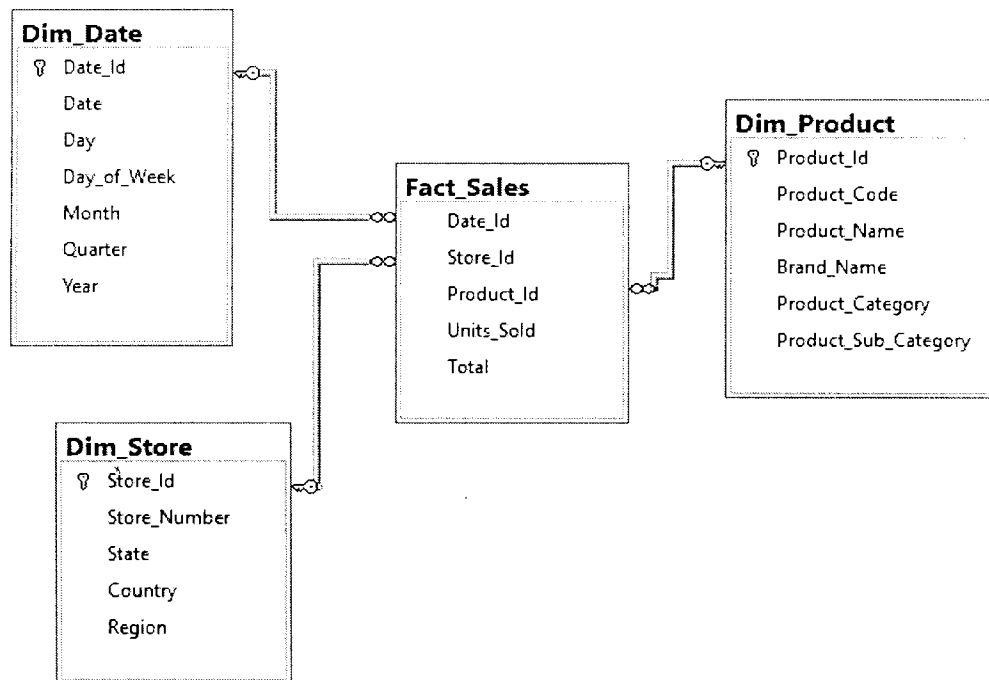
1. List four goals of a Data Warehousing & Business Intelligence solutions. (4 marks)
2. Compare a Business Intelligence solution and a Data Warehouse. (4 marks)
3. Design a comprehensive Business Intelligence solution architecture for an organization using following components.
 - Data sources
 - ETL
 - Data Warehouse
 - Semantic Layer/OLAP Server
 - Visualization(5 marks)
4. List available Microsoft tools to develop each of the following components.
 - a. ETL
 - b. Staging Layer
 - c. Data Warehouse
 - d. OLAP Cubes
 - e. Reports/Visualizations(5 marks)

Question 2**(30 marks)**

1. Differentiate Star schema and Snowflake schema. (4 marks)
2. Model a Snowflake schema for the given Star schema design below. (6 marks)

Note:

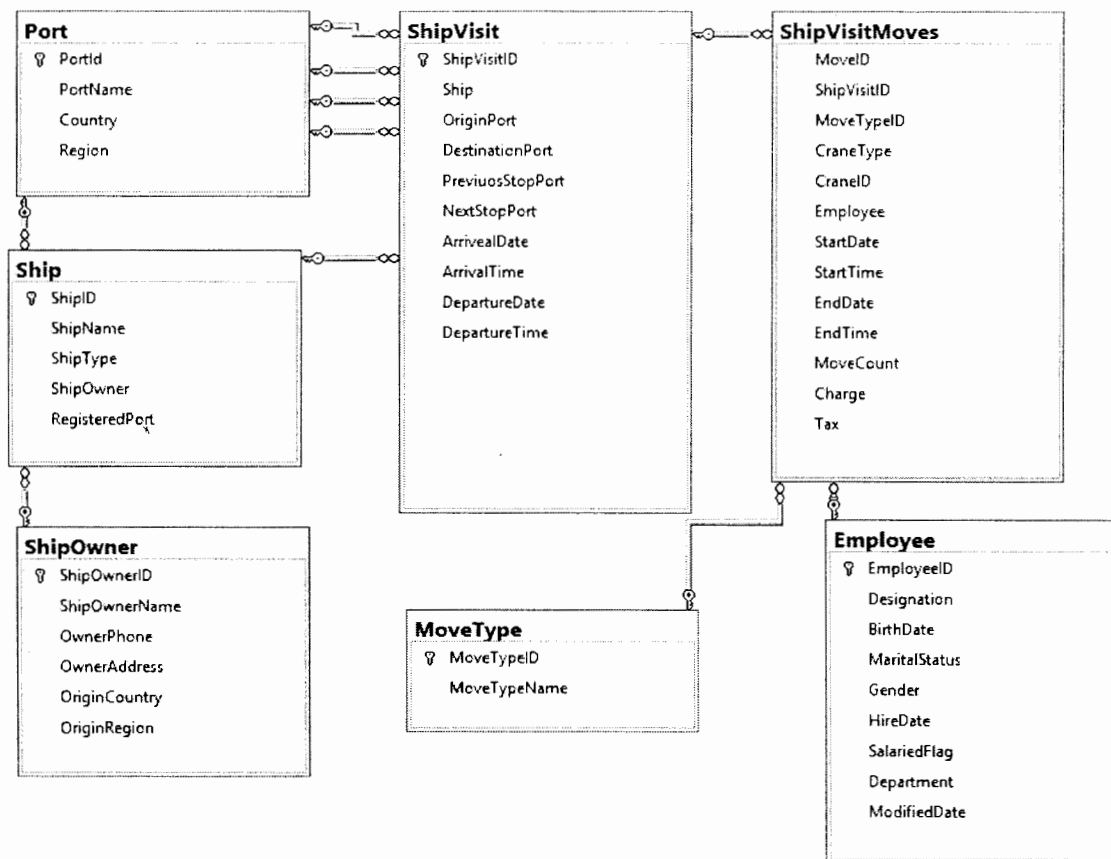
- One Product Category contains many Product Sub Categories, and one sub category contains many Products in Dim_Products.
- Product Category and Brand are mutually exclusive (no relationship) in Dim_Products.
- One Region contains many Countries, one Country contains many States, and one State contains many Stores.



3. Consider the below ER diagram of an OLTP database which stores data related to a terminal operations in a local port run by the ABC Company. The company operates cranes to move containers from one ship to another. Below is a brief description of their OLTP database.

Information:

- **ShipOwner** table stored details of companies which owns ships.
- **Port** table holds details of well-known ports around the world.
- **Ship** table holds details of ships.
 - o **ShipOwner** column is a foreign key to the **ShipOwnerID** of **ShipOwner** table
 - o **RegisteredPort** column is a foreign key to the **PortID** of **Port** table
- **ShipVisit** table holds the details about visits of ships to the terminals operated by the ABC Company.
 - o **Ship** column is a foreign key to the **ShipID** of **Ship** table
 - o **OriginPort** column is a foreign key to the **PortID** of **Port** table
 - o **DestinationPort** column is a foreign key to the **PortID** of **Port** table
 - o **NestStopPort** column is a foreign key to the **PortID** of **Port** table
 - o **PreviousStopPort** column is a foreign key to the **PortID** of **Port** table
- **Employee** table holds details of employees
- **MoveType** table holds details of different crane move types
- **ShipVisitMoves** table holds details of crane moves. An employee operates a crane, and move containers from one location to another. For the same type of moves made, in a single session, from/to a specific ship visit, is stored as a single row.
 - o **ShipVisitID** column is a foreign key to the **ShipVisitID** of **ShipVisit** table
 - o **Employee** column is a foreign key to the **EmployeeID** of **Employee** table
 - o **MoveTypeID** column is a foreign key to the **MoveTypeID** of **MoveType** table.

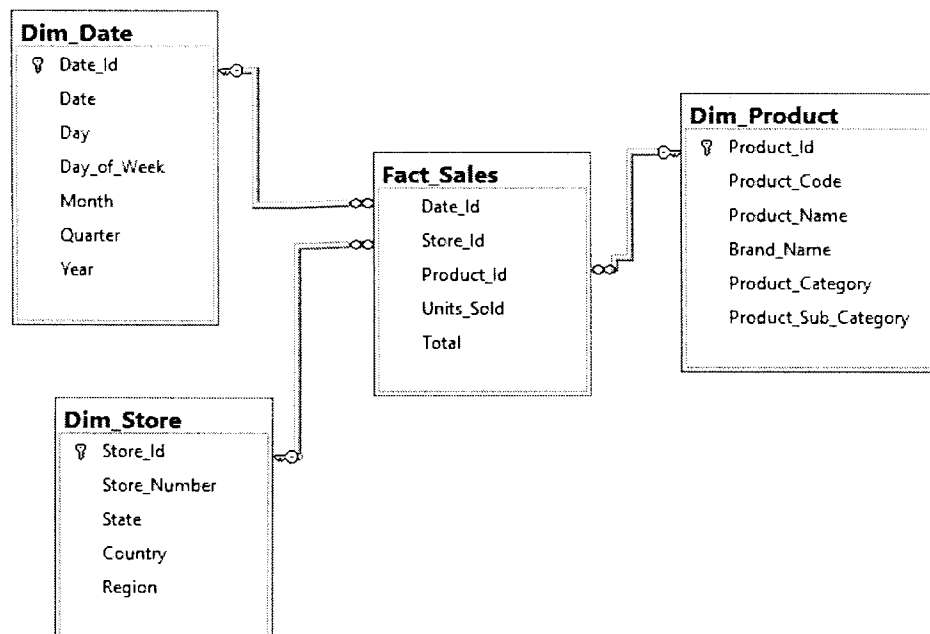


- Design a suitable dimensional model (star or snowflake) schema for a data warehouse for ABC Company. State any assumptions you made for your model. Please note that the Department column of Employee table should be maintained for historical data analysis. (18 marks)
- Recommend two (2) hierarchies that can be derived in the data warehouse design? You may consider hierarchies which are not there in the OLTP system, but can be derived in a Data warehouse design for this scenario. (2 marks)

Question 3

(15 marks)

1. Explain what “Change Data Capture” is. (3 marks)
2. Compare different data loading mechanisms (how history is managed) when loading data into following types of Slowly Changing Dimensions:
 - a. Type 1 SCD
 - b. Type 2 SCD
 - c. Type 3 SCD(6 marks)
3. Consider the following Data Warehouse design. You need to model an ETL component to load data from source system (assume only one source system is available and all required data are extracted from the said source system) to the Data Warehouse.



When loading data to each of the above tables, state your recommendations considering the following aspects.

- a. Data loading sequence (order of tables in which data will be loaded)
- b. Possible transformations (for the fact table)
- c. Possible data cleansing required (for Product and Store dimensions)

(6 marks)

Question 4**(18 marks)**

1. Differentiate OLAP storage types ROLAP, MOLAP and HOLAP in terms of following aspects:
 - a. How it is implemented
 - b. Advantages
 - c. Disadvantages(8 marks)
2. Compare OLAP operations Roll-up and Drill-down (2 marks)
3. Compare OLAP operations Slice and Dice (2 marks)
4. Implementation of OLAP Cubes require additional investment since the implementation requires proprietary technologies. Assuming you are a BI solution architect that design & develops BI solutions for a large company which generates millions of data daily and contains many departments, justify the requirement of investing in such technologies to the top management. (6 marks)

Question 5**(19 marks)**

1. Briefly explain functionalities of following SQL Server Reporting Service (SSRS) Components:
 - a. Report Builder
 - b. Mobile Report Publisher
 - c. SSRS Web Portal
 - d. Report Server Database(8 marks)
2. Differentiate following types of SSRS reports:
 - a. Paginated reports
 - b. Mobile reports
 - c. KPIs(3 marks)
3. Differentiate "Cached Reports" and "Snapshots" (4 marks)
4. Explain the use of 'Roles and Permissions' used in SSRS (4 marks)